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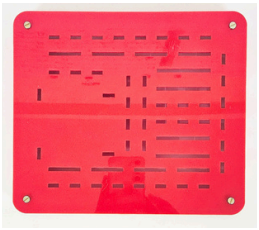
SESSION NAME

Session

4

STRUCTURE AND FASTENERS OF MICROCONTROLLERS

1



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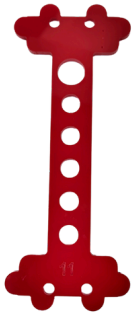


1

Spacers

2

L Clamp

3

Spacer Screws

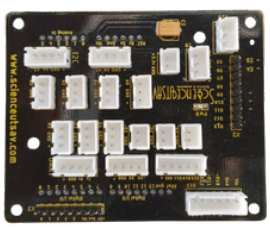
4

Nuts

5

Screw

6

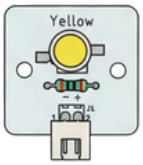


Arduino With Shield

7

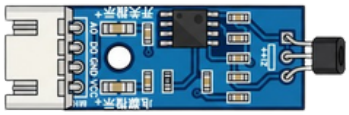
Arduino cable

ELECTRONIC COMPONENTS



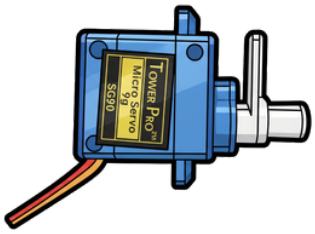
1 Watt LED PCB- 1 Watt LED is a high-brightness light-emitting diode that produces much stronger light compared to normal small LEDs.

DIP LED- A DIP LED (Dual In-line Package LED) is a type of LED with two long metal pins used for easy connection on breadboards and PCB circuits.



Magnetic sensor- A Magnetic Sensor is an electronic sensor that detects the presence or absence of a magnetic field. It is commonly used to sense the opening and closing of doors, count rotating objects, or detect nearby magnets. The sensor outputs a signal when a magnet comes close, making it useful in security systems, automation, and robotics.

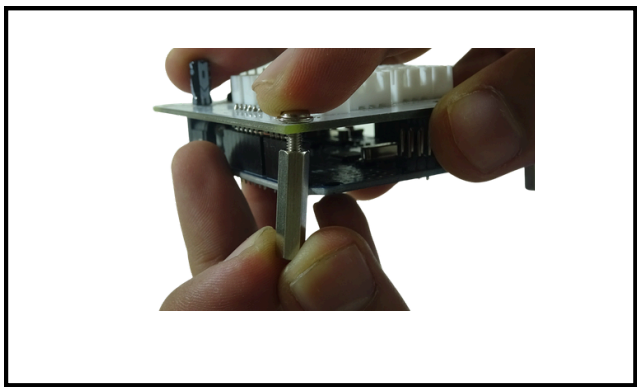
DC motor- A DC Motor is an electric motor that converts direct current (DC) electrical energy into mechanical rotational motion. It is widely used in projects and devices to rotate wheels, fans, gears, and other moving parts. DC motors are popular because they are simple to use, compact, and provide controllable speed and direction.

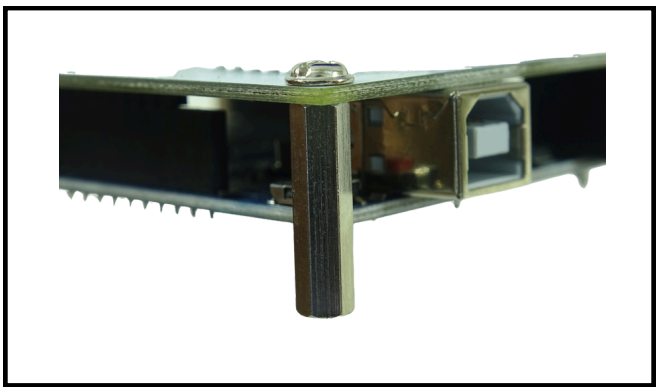


Servo Motor- A servo motor is a compact motor that rotates to a specific angle with high precision. It is commonly used in robotics and automation for accurate position control.

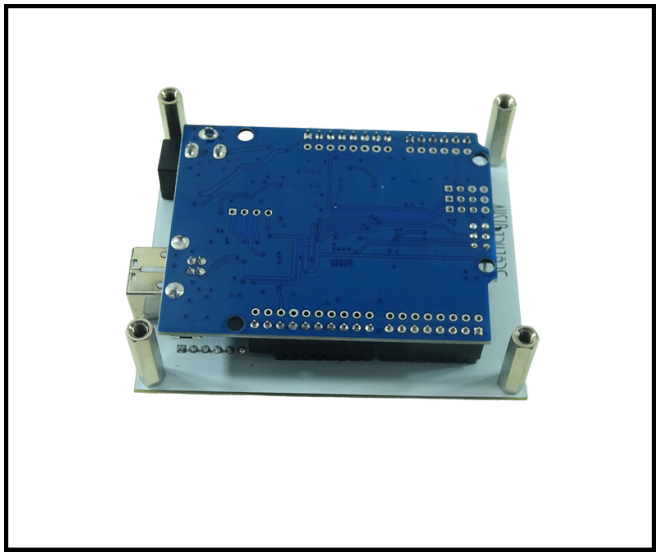
Procedure :

Step 1 : Fit the spacer to Arduino Shield as shown below using Spacer screw

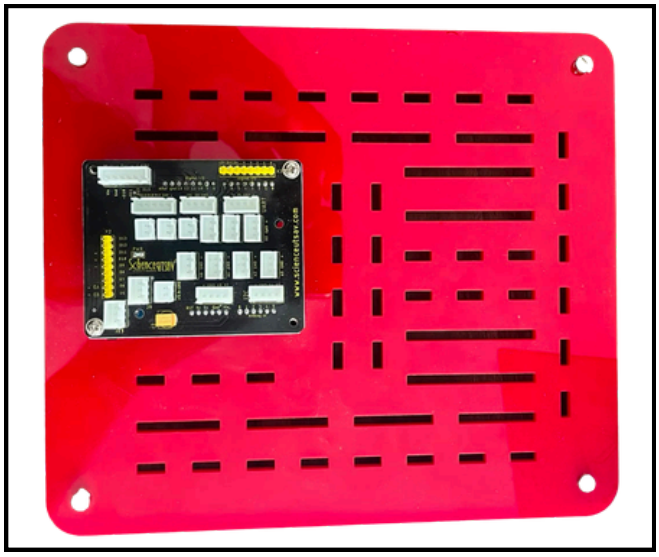




Step 2 : Repeat the step 1 for all 4 corners of Arduino Shield



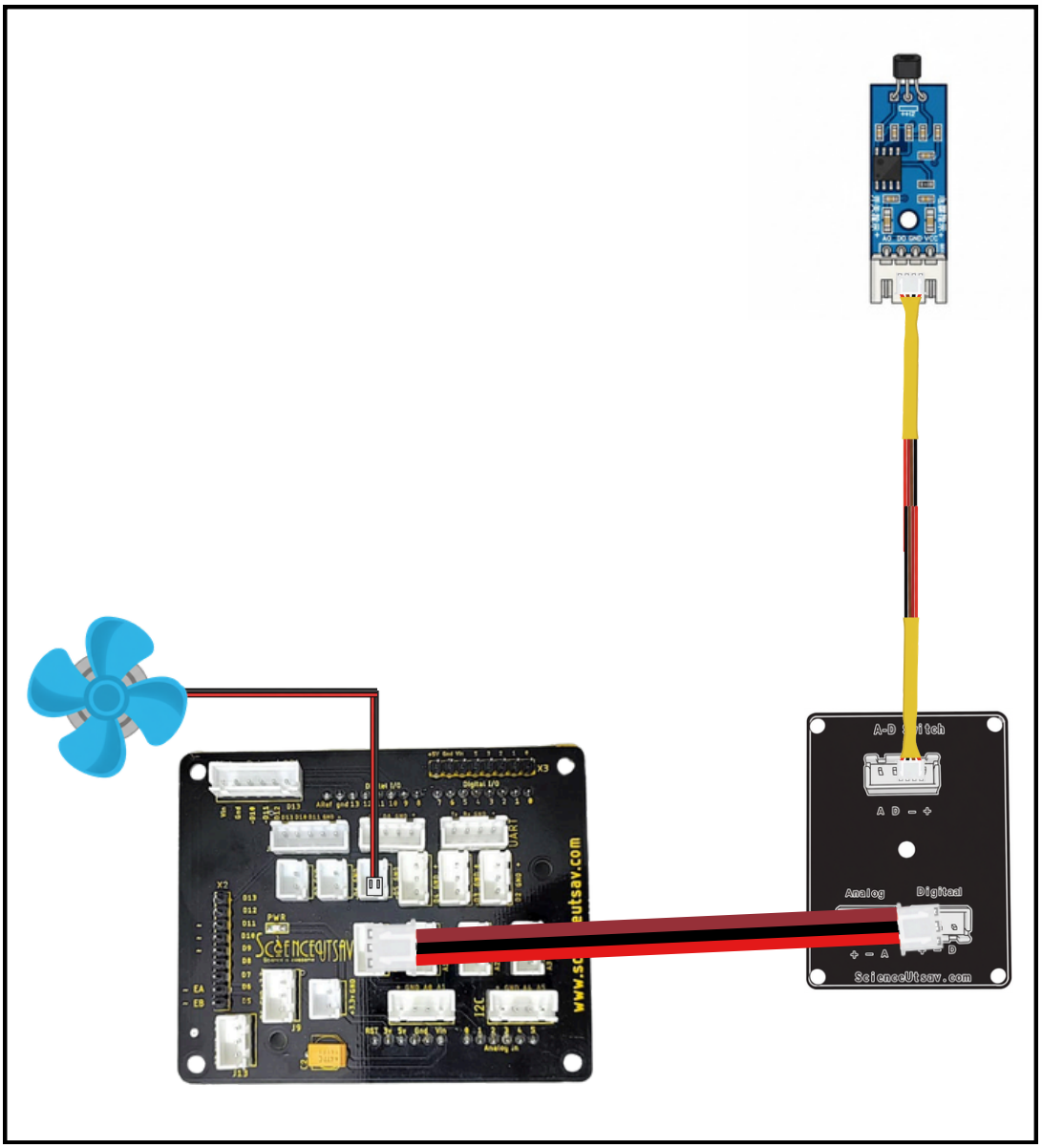
Step 3 : Place the Arduino Shield to the base board as shown below



Step 4 : Fix it with the spacer screw from the bottom as shown below

BEGINNER PROJECT

The circuit diagram of the project is given below



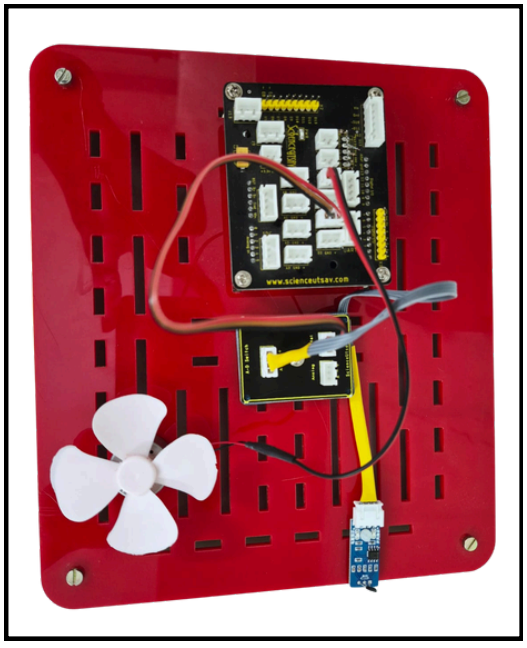
CONNECTION:

1. DC motor- D5
2. AD switch- A0

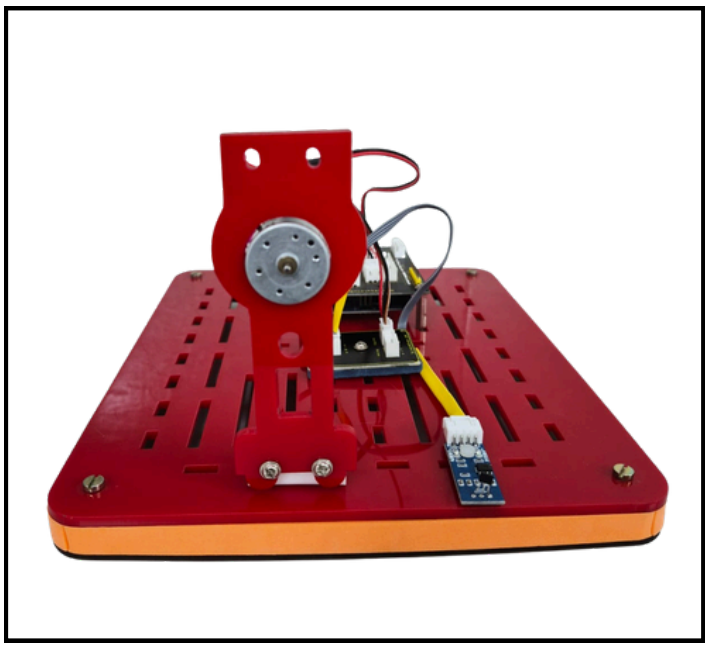
Step 1- Secure the AD switch PCB to the Pegboard

Step 2- Connect the magnetic sensor to the AD switch PCB as shown

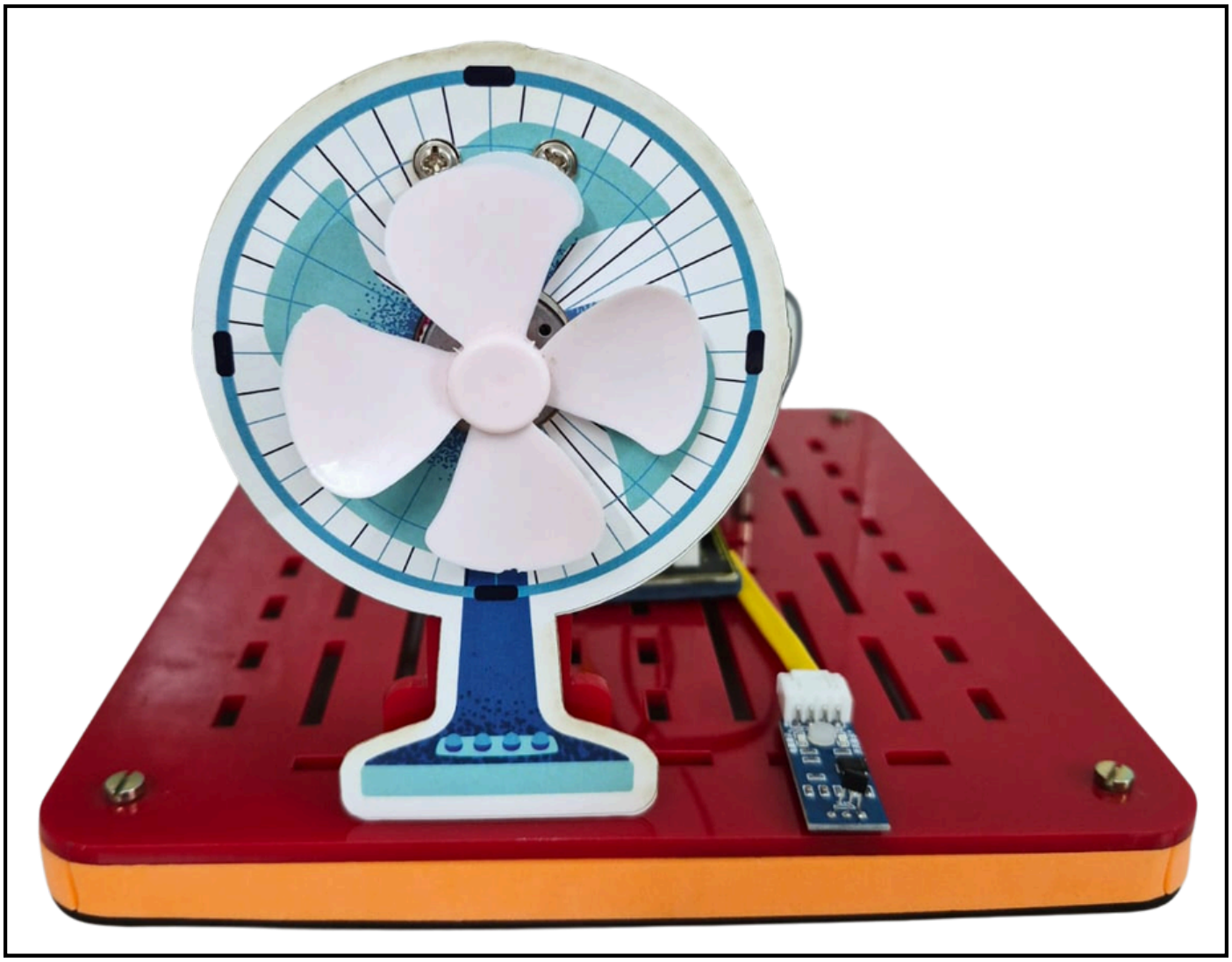
Step 3- Attach the DC motor to the shield



Step 4- Fix the DC motor holder to the board and attach the motor



Step 5- Attach the template to the stand and fix the fan to the motor

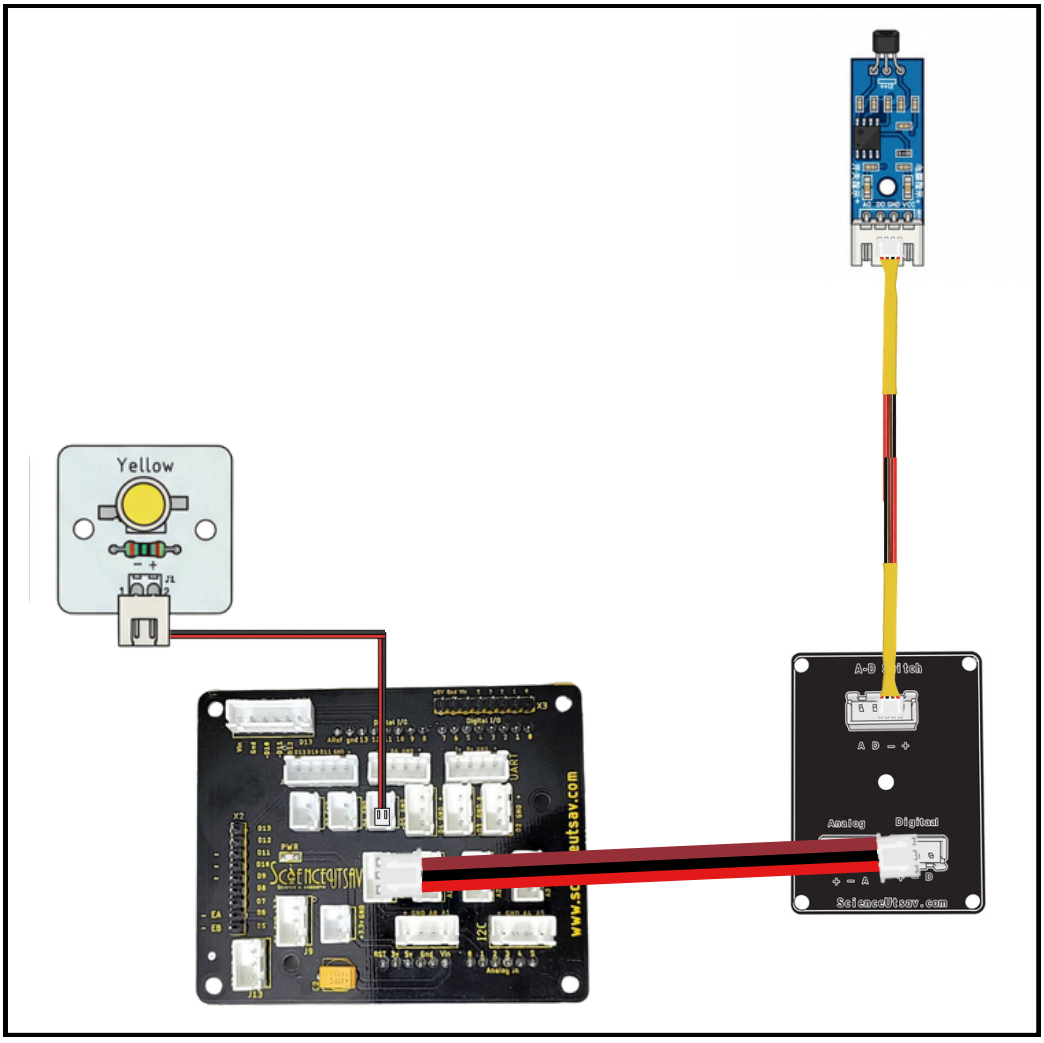


Code

```
const int magneticSensor = A0;
const int motor = 5;

const int MAGNET_DETECTED = LOW; // change
to HIGH if opposite

void setup() {
  pinMode(magneticSensor, INPUT);
  pinMode(motor, OUTPUT);
}

void loop() {
  if (digitalRead(magneticSensor) ==
MAGNET_DETECTED) {
    digitalWrite(motor, HIGH);
  } else {
    digitalWrite(motor, LOW);
  }
}
```

BEGINNER PROJECT

The circuit diagram of the project is given below



CONNECTION:

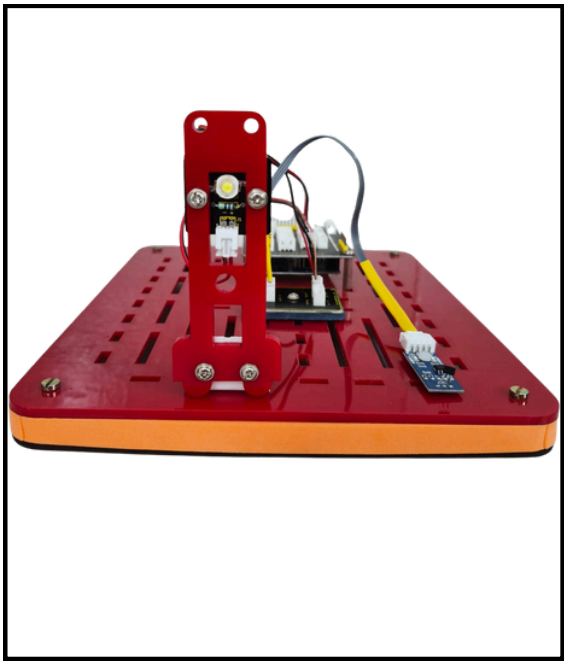
1. 1W LED- D5
2. AD switch- A0

Step 1- Secure the AD switch PCB to the Pegboard

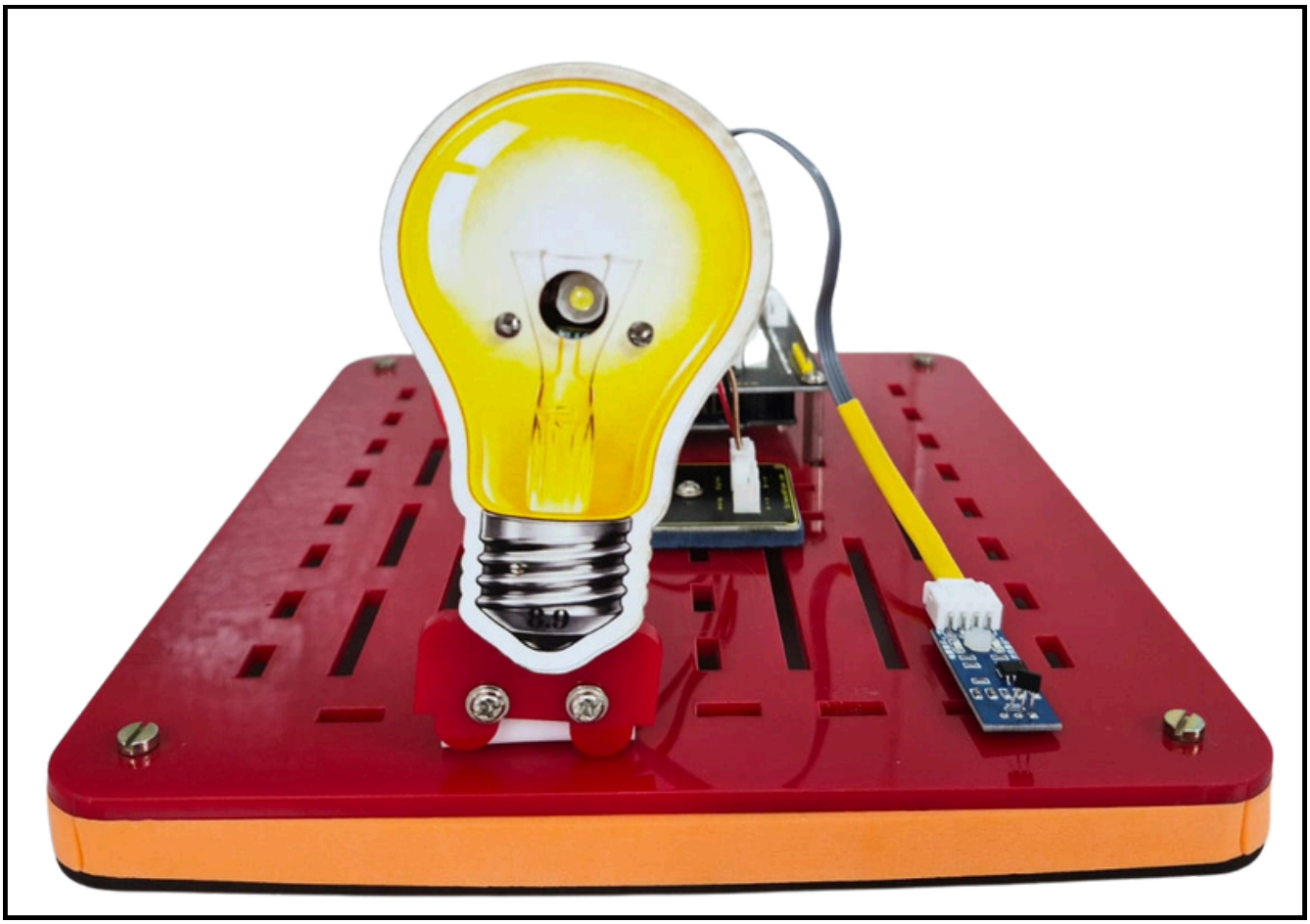
Step 2- Connect the magnetic sensor to the AD switch PCB as shown

Step 3- Attach the 1W LED to the shield

Step 4- Fix th stand to the board and attach the LED to it



Step 5- Attach the template to the stand

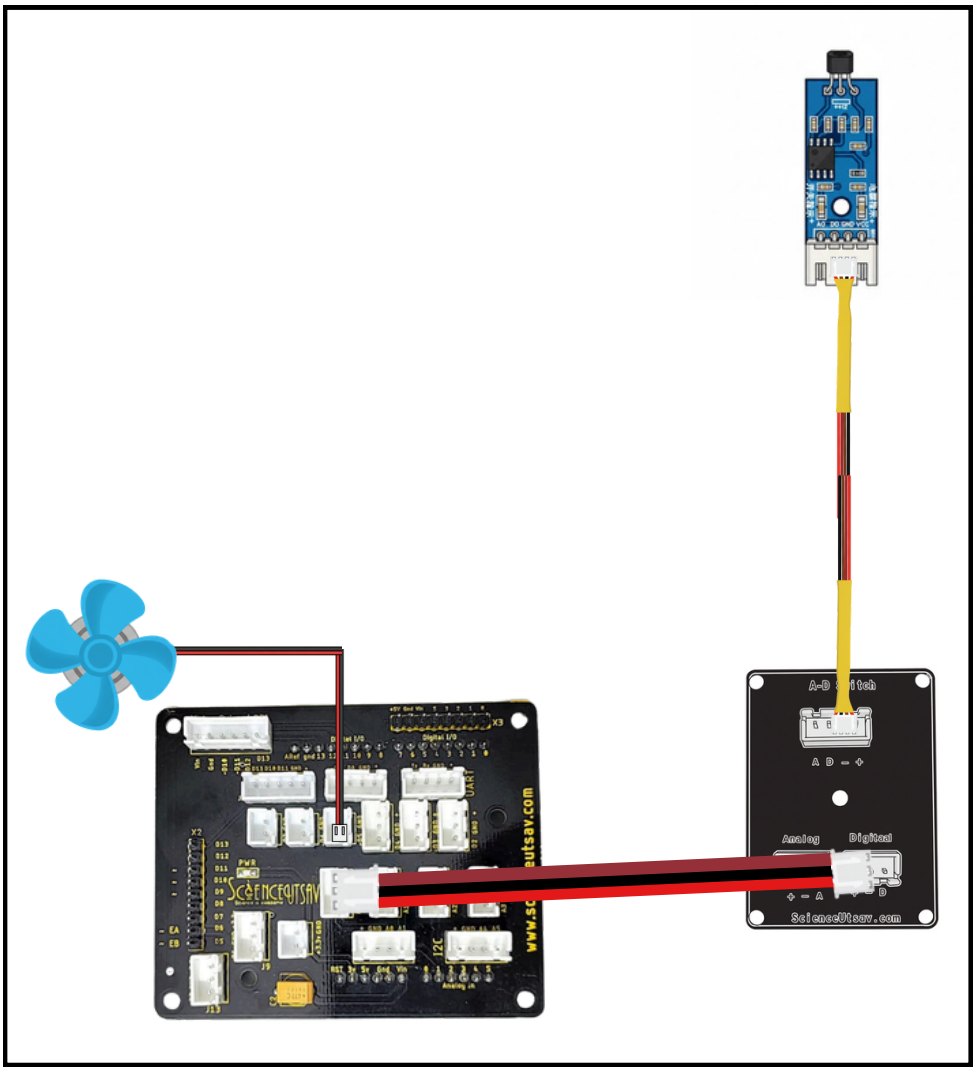


Code

```
const int magneticSensor = A0;
const int motor = 5;

const int MAGNET_DETECTED = LOW; // change
to HIGH if opposite

void setup() {
  pinMode(magneticSensor, INPUT);
  pinMode(motor, OUTPUT);
}

void loop() {
  if (digitalRead(magneticSensor) ==
MAGNET_DETECTED) {
    digitalWrite(motor, HIGH);
  } else {
    digitalWrite(motor, LOW);
  }
}
```

INTERMEDIATE PROJECT

The circuit diagram of the project is given below



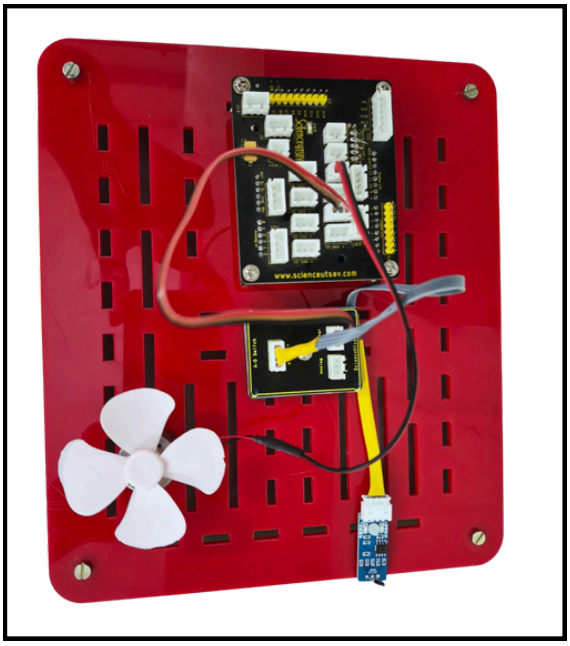
CONNECTION:

1. DC motor- D5
2. AD switch- A0

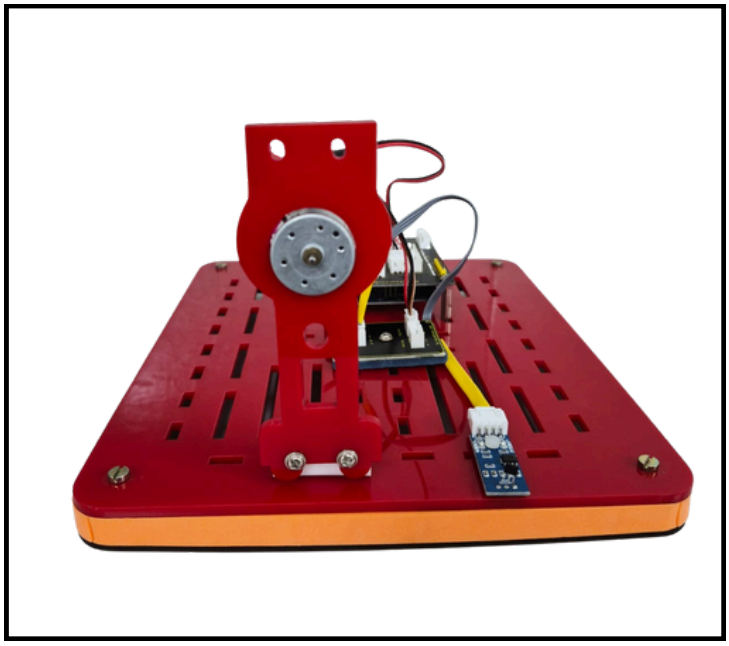
Step 1- Secure the AD switch PCB to the Pegboard

Step 2- Connect the magnetic sensor to the AD switch PCB as shown

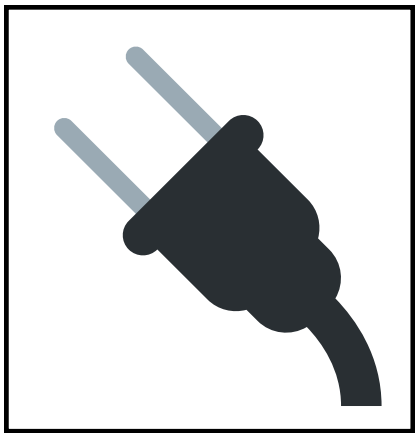
Step 3- Attach the DC motor to the shield



Step 4- Fix the DC motor holder to the board and attach the motor



Step 5- Attach the magnet to the back of the template as shown



Step 6- Attach both the template to the stands as shown

Step 7- Fix a pulley to the small template and attach it to the motor

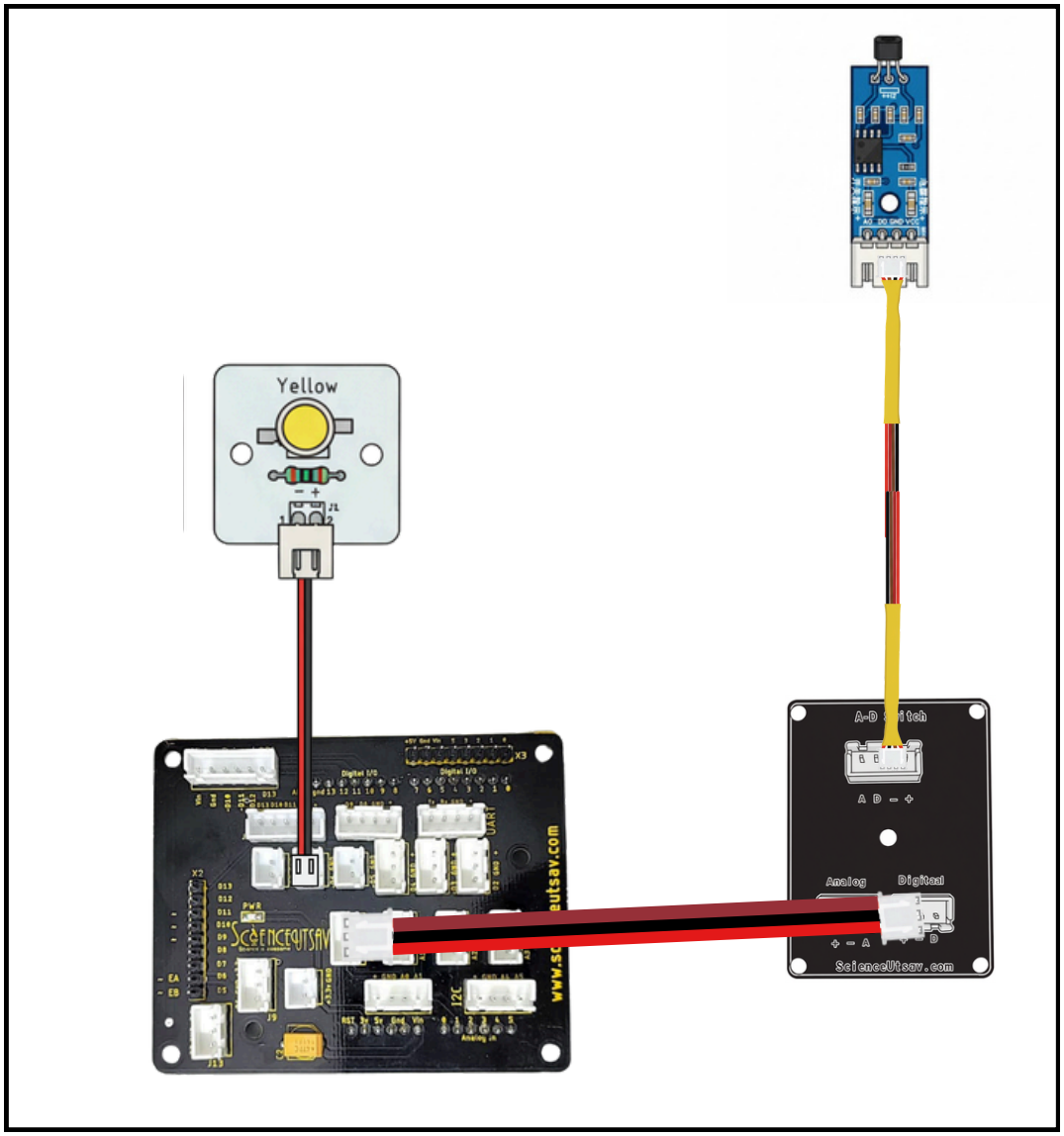
Step 8- Bring the magnet near the magnetic sensor to play with your project

Code

```
// Session 4 Grade 7
// Washing Machine
// Plug with magnet near socket sensor = motor
rotates
// Magnetic sensor used as DIGITAL sensor on A0

const int magneticSensor = A0;
const int motor = 5;

const int MAGNET_DETECTED = LOW;  // change to
HIGH if opposite

void setup() {
  pinMode(magneticSensor, INPUT);
  pinMode(motor, OUTPUT);
}

void loop() {
  if (digitalRead(magneticSensor) ==
MAGNET_DETECTED) {
    digitalWrite(motor, HIGH);
  } else {
    digitalWrite(motor, LOW);
  }
}
```

INTERMEDIATE PROJECT

The circuit diagram of the project is given below



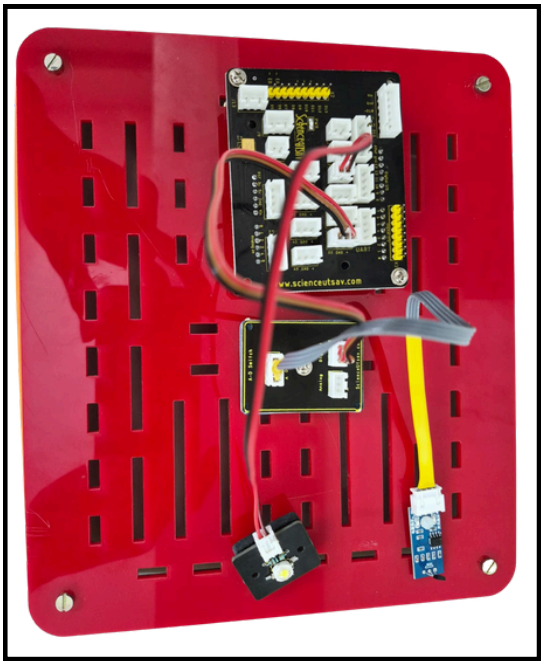
CONNECTION:

1. 1W LED- D6
2. AD switch- A0

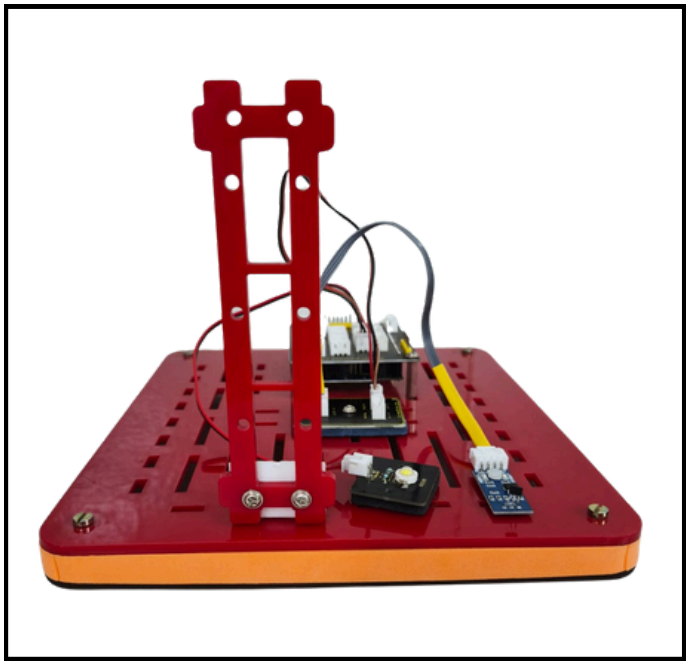
Step 1- Secure the AD switch PCB to the Pegboard

Step 2- Connect the magnetic sensor to the AD switch PCB as shown

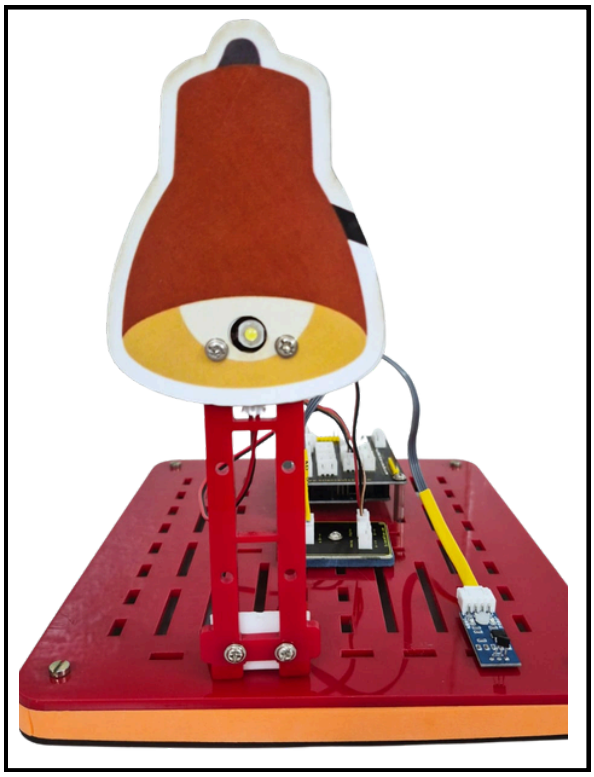
Step 3- Attach the 1W LED to the shield



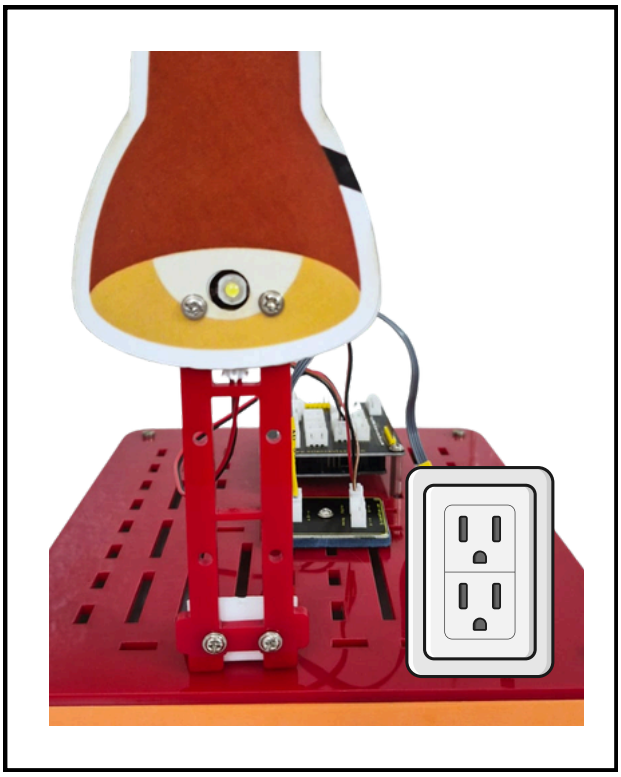
Step 4- Fix the stand to the board as shown



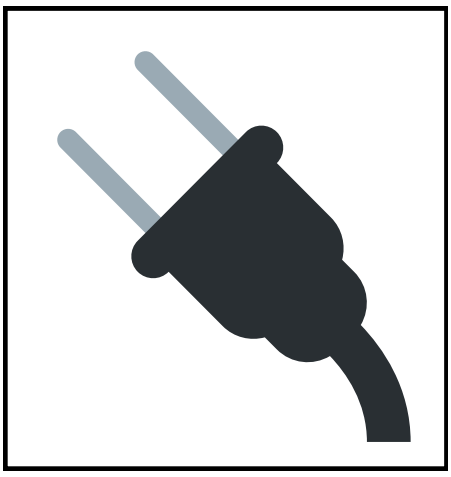
Step 5- Attach the template and fix the LED to it



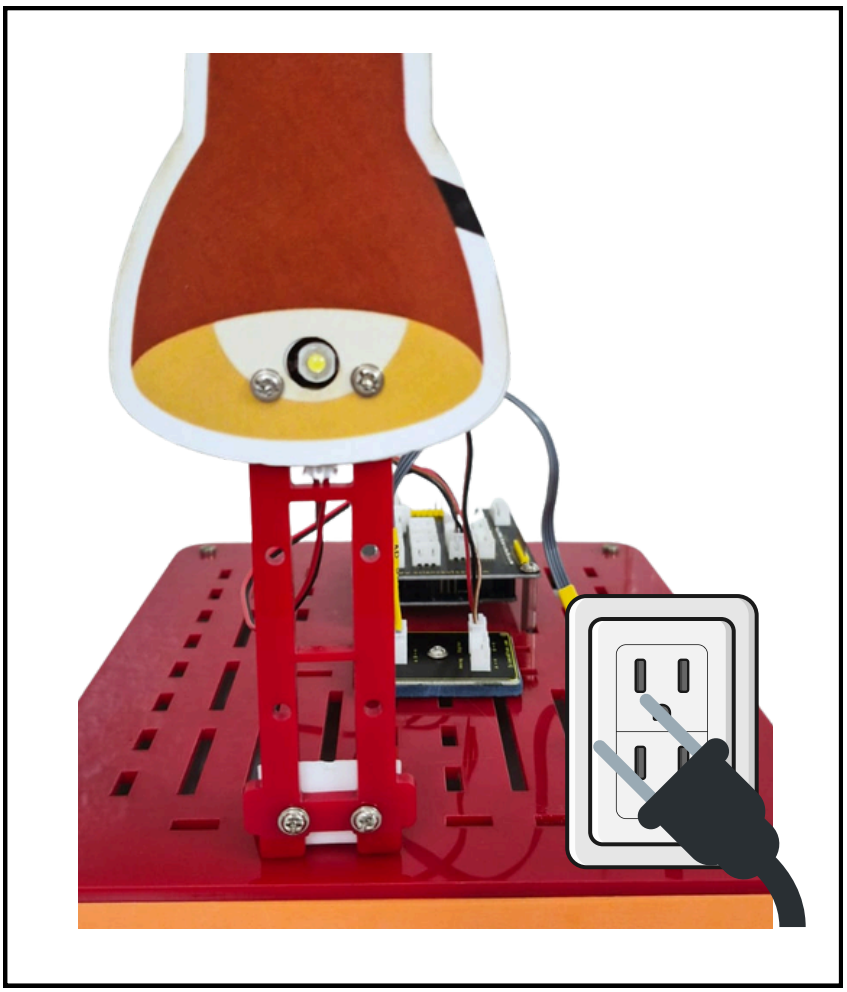
Step 6- Attach the other template as shown



Step 7- Attach the magnet to the back of the template as shown



Step 8- Bring the magnet near the magnetic sensor to play with your project

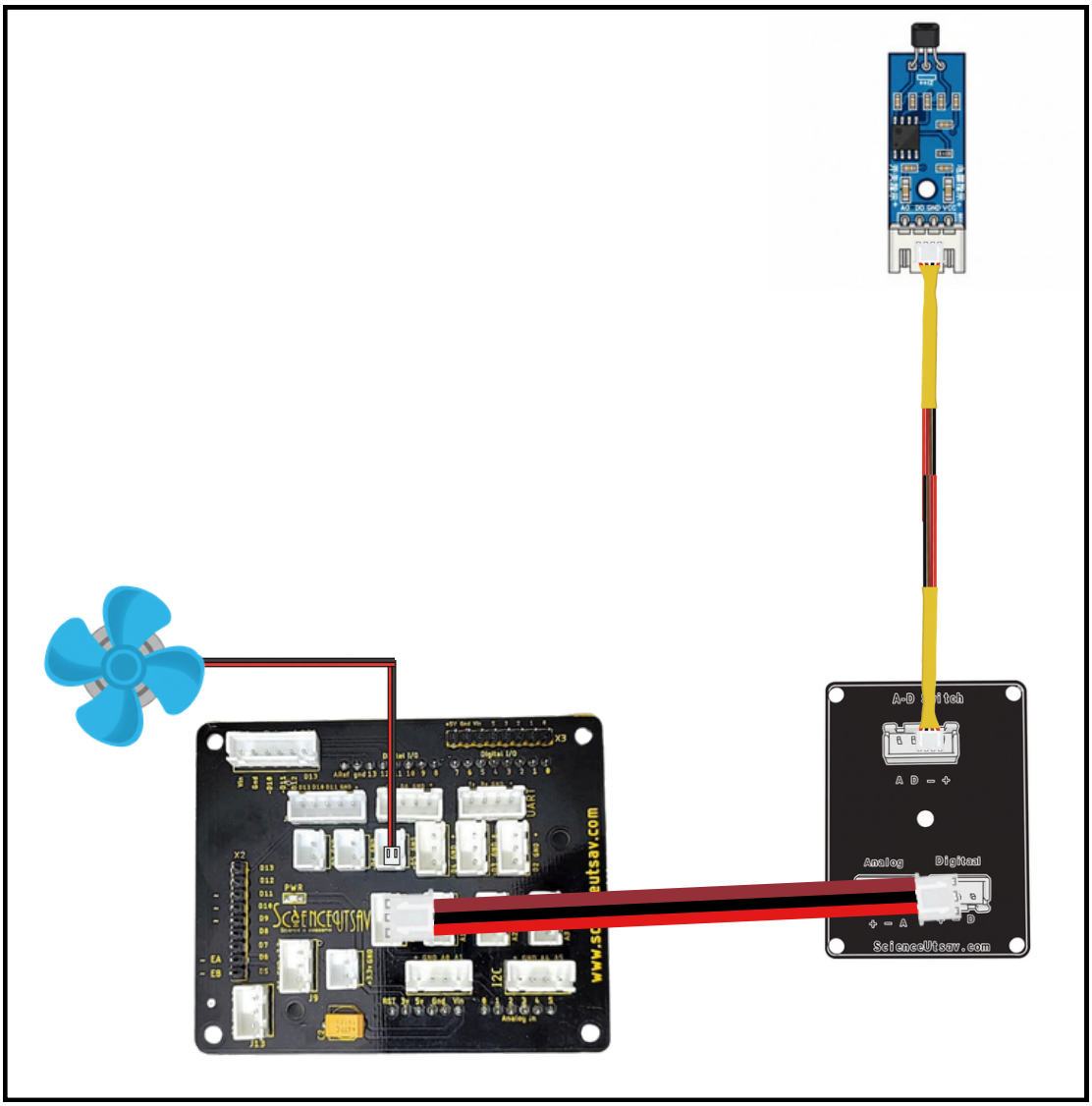


Code

```
// Session 4 Grade 7
// Magnetic Table Lamp
// Plug with magnet near socket sensor = LED ON
// Magnetic sensor used as DIGITAL sensor on A0

const int magneticSensor = A0;
const int led = 6;

const int MAGNET_DETECTED = LOW;  // change to HIGH if
opposite

void setup() {
  pinMode(magneticSensor, INPUT);
  pinMode(led, OUTPUT);
}

void loop() {
  if (digitalRead(magneticSensor) == MAGNET_DETECTED) {
    digitalWrite(led, HIGH);
  } else {
    digitalWrite(led, LOW);
  }
}
```

ADVANCE PROJECT

The circuit diagram of the project is given below



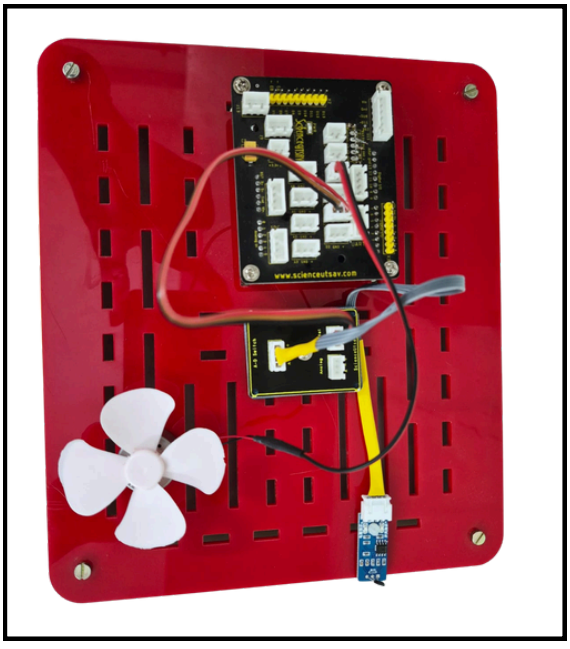
CONNECTION:

1. DC motor- D5
2. AD switch- A0

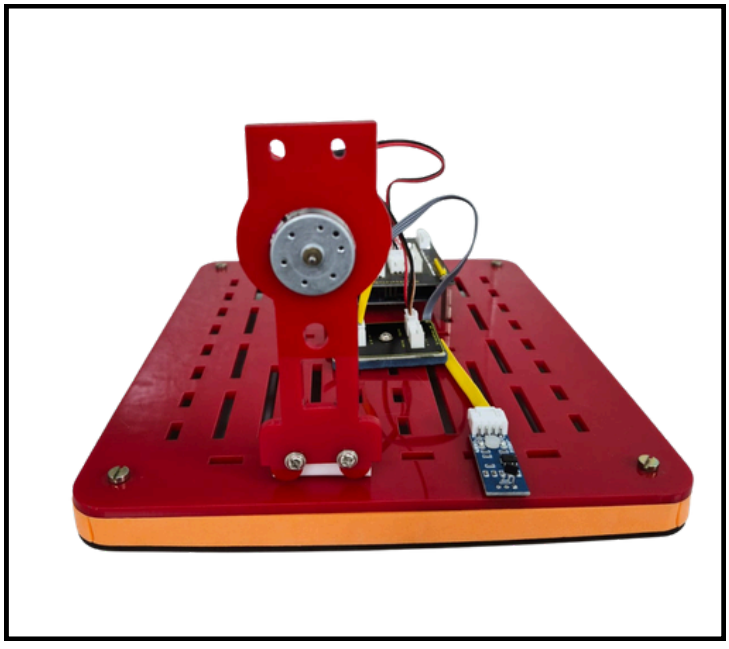
Step 1- Secure the AD switch PCB to the Pegboard

Step 2- Connect the magnetic sensor to the AD switch PCB as shown

Step 3- Attach the DC motor to the shield



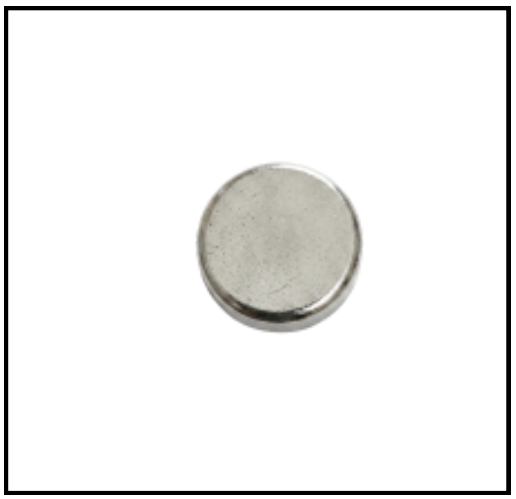
Step 4- Fix the DC motor holder to the board and attach the motor



Step 5- Attach the template to the stand. Fix the pulley to the wheel and attach it to the motor

Step 6- Fix the magnetic sensor to the back of the template as shown

Step 7- Fix the magnet to the back of the template as shown



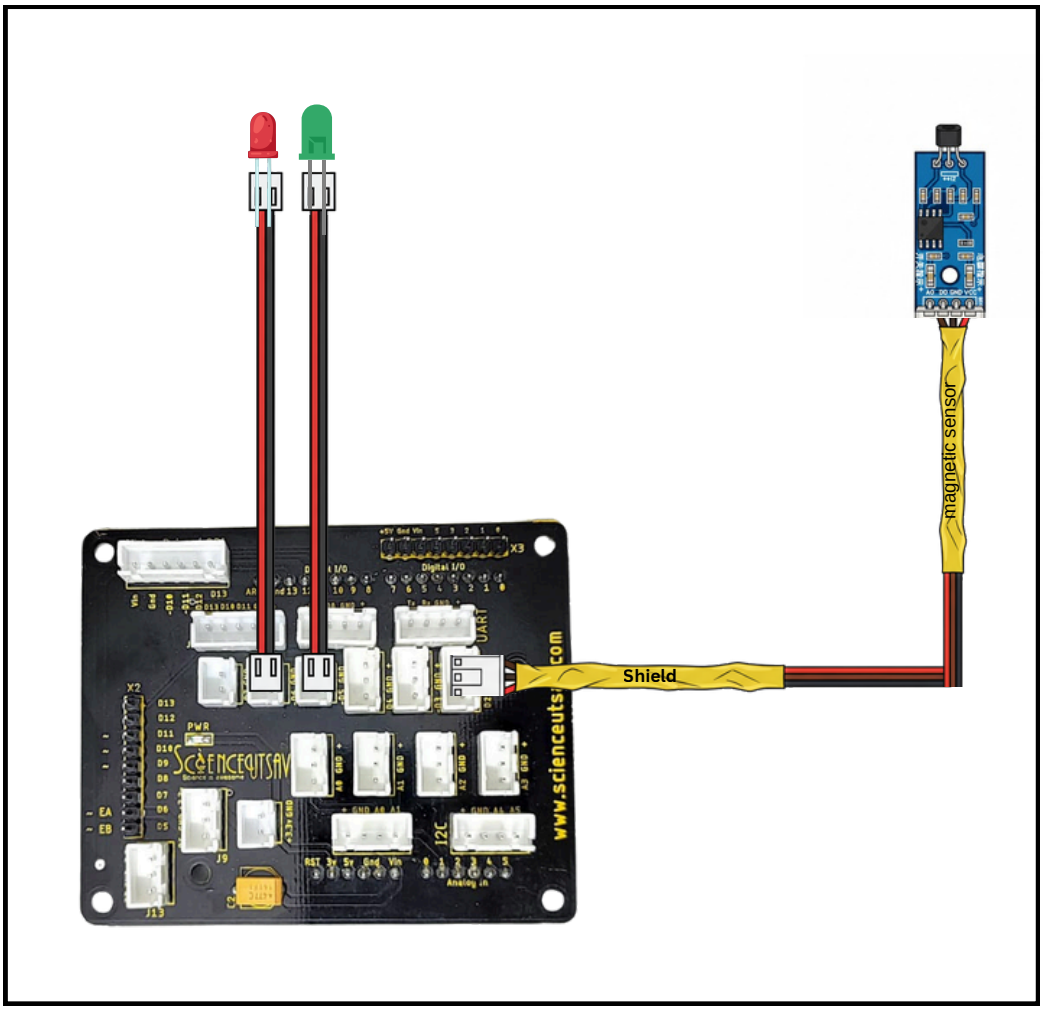
Step 8- Bring the magnet near the magnetic sensor to play with your project

Code

```
// Session 4 Grade 8
// Helmet Bike
// Helmet magnet detected = bike wheel motor ON
// Magnetic sensor used as DIGITAL sensor on A0

const int magneticSensor = A0;
const int motor = 5;

const int MAGNET_DETECTED = LOW;  // change to
HIGH if opposite

void setup() {
  pinMode(magneticSensor, INPUT);
  pinMode(motor, OUTPUT);
}

void loop() {
  if (digitalRead(magneticSensor) ==
MAGNET_DETECTED) {
    digitalWrite(motor, HIGH);
  } else {
    digitalWrite(motor, LOW);
  }
}
```

ADVANCE PROJECT

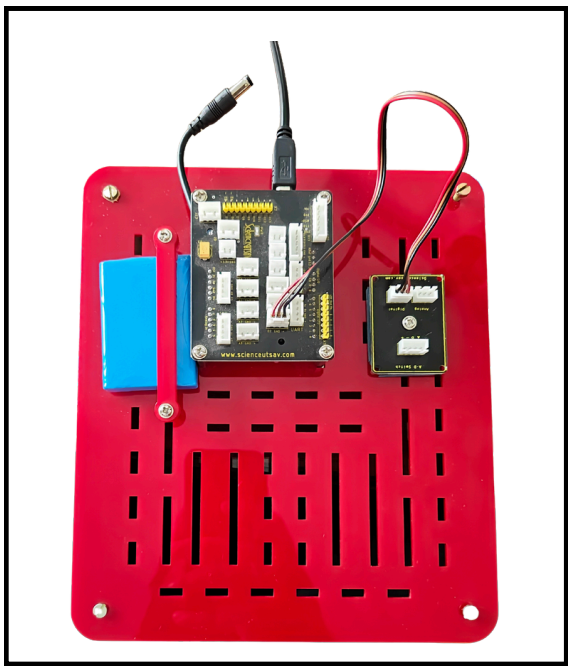
The circuit diagram of the project is given below



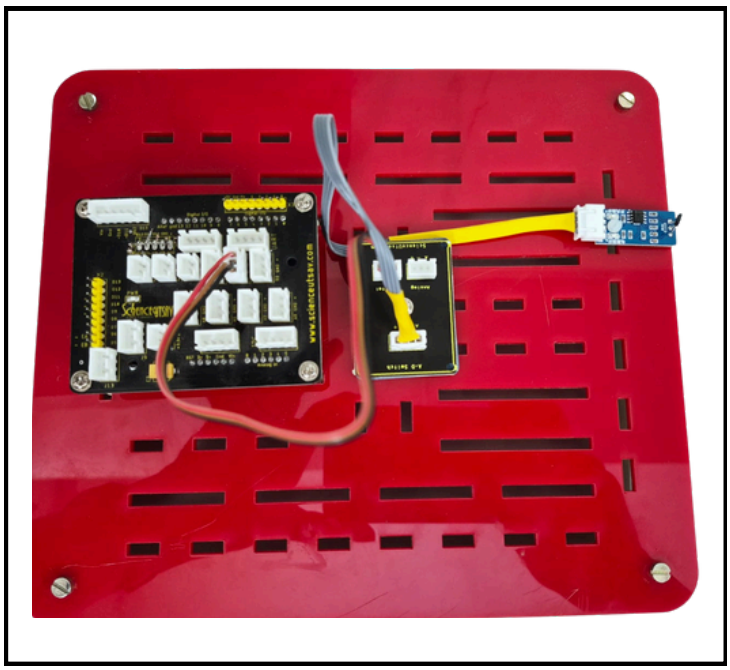
CONNECTION:

1. AD switch- A0
2. Green LED- D5
3. Red LED- D6

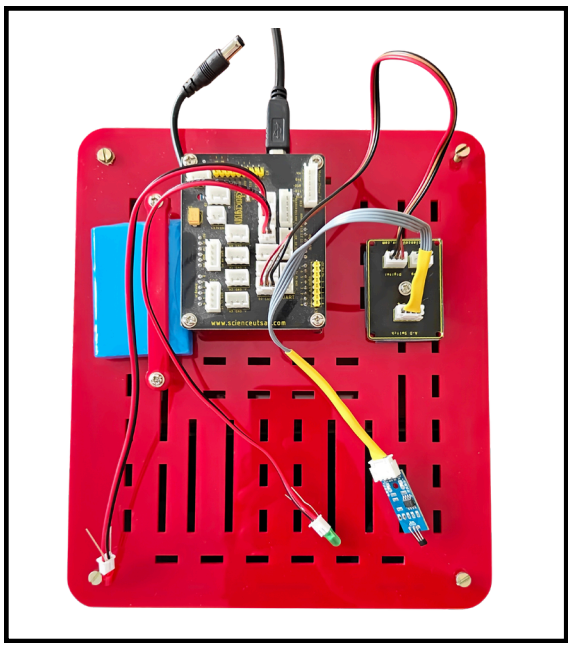
Step 1- Secure the AD switch PCB to the Pegboard



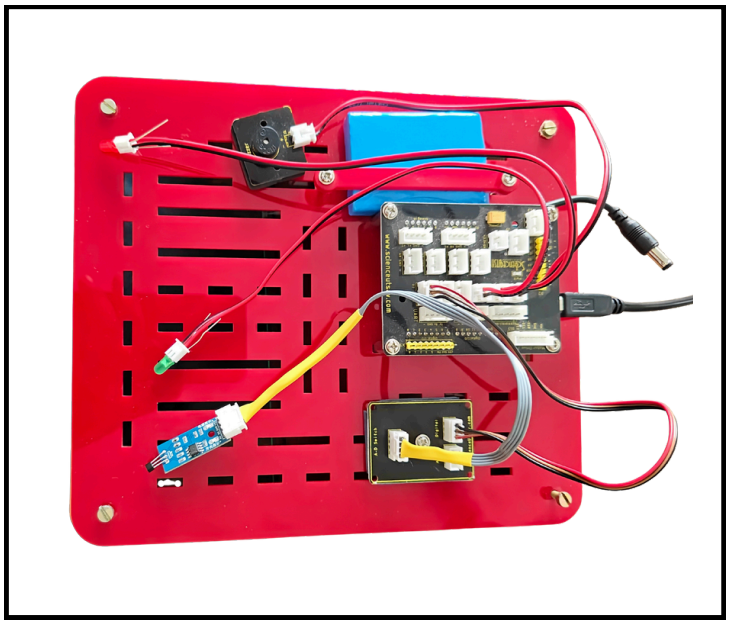
Step 2- Connect the magnetic sensor to the AD switch PCB as shown



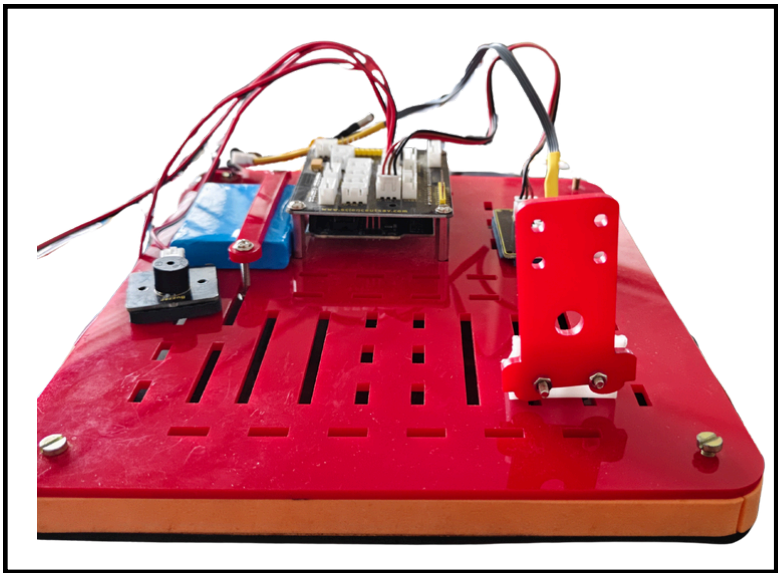
Step 3- Attach the red and green leds to the shield



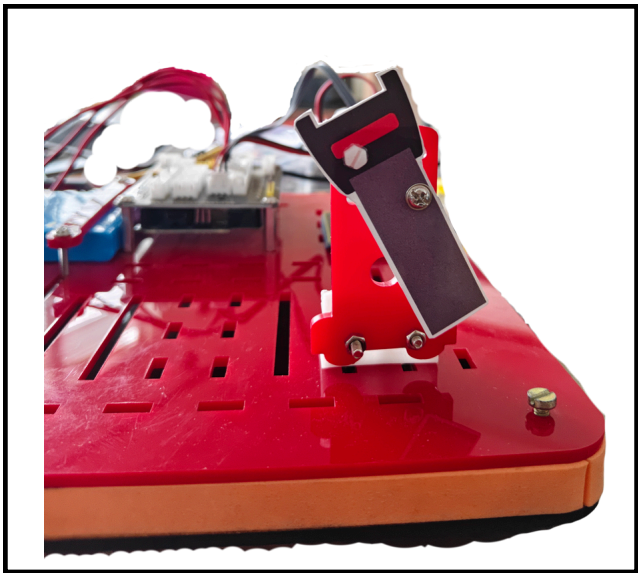
Step 4- Attach the buzzer



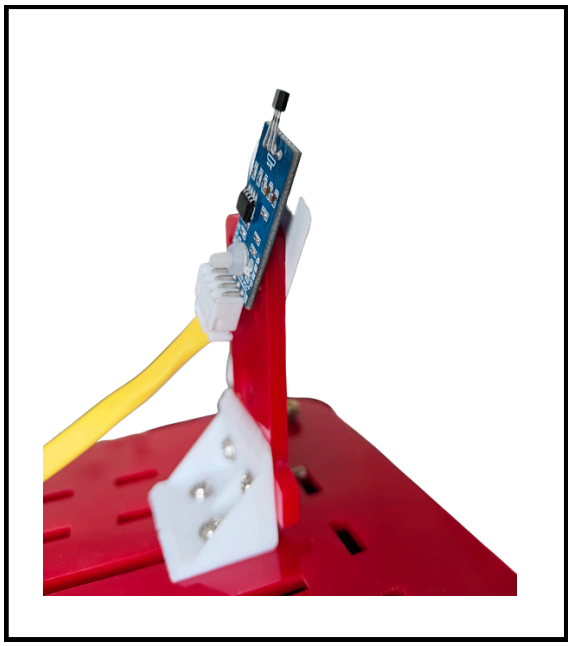
Step 5- Secure the stand to the Pegboard as shown



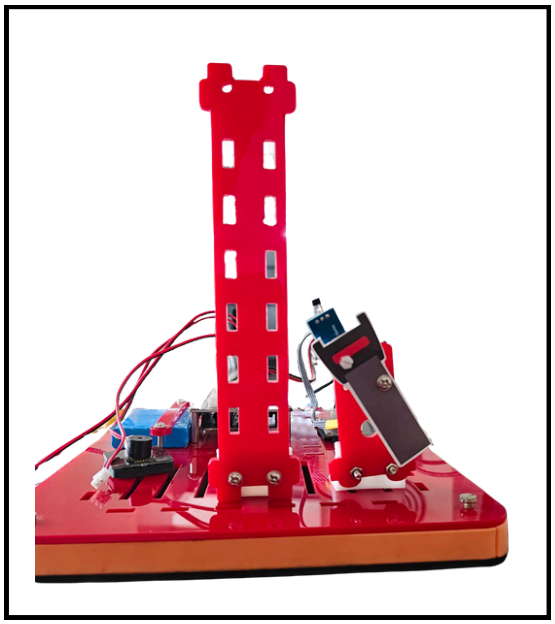
Step 6- Attach the template to the stand



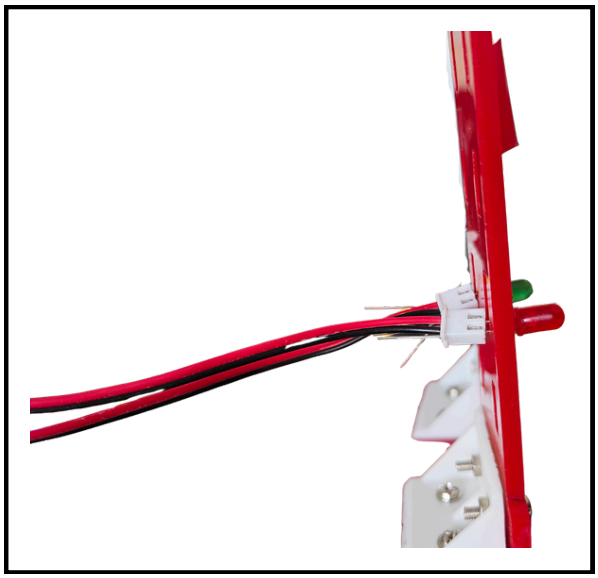
Step 7- Screw in the halleffect sensor to the back of the template



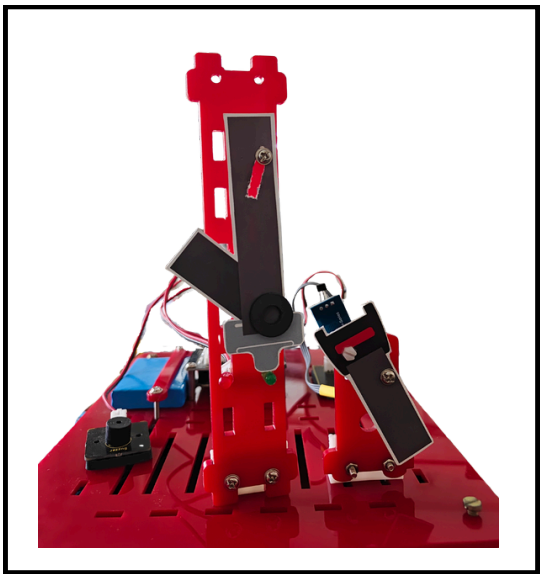
Step 8- Fix the stand as shown



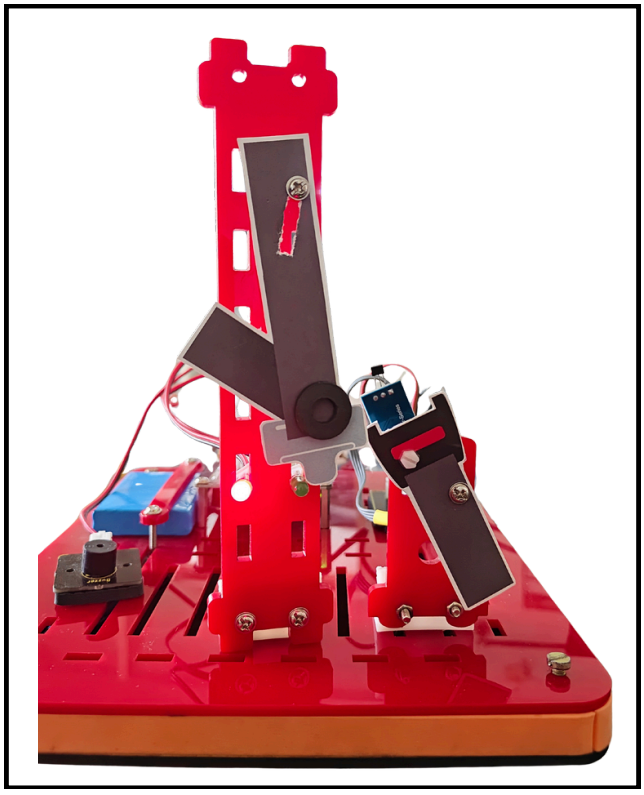
Step 9- Attach the leds as shown

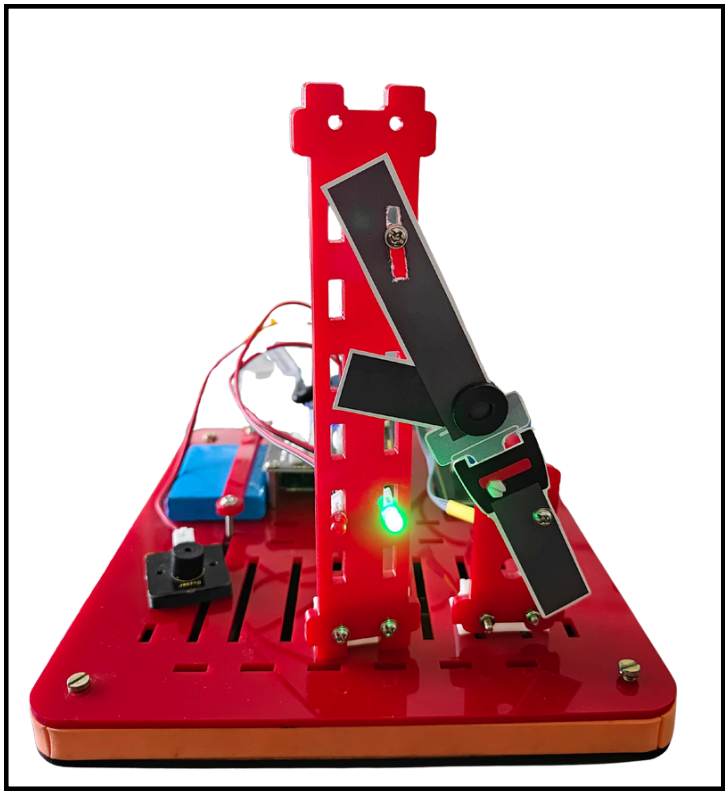


Step 10- Attach the template to the stand as shown and put the magnet as well



Step 11- Upload the code and observe



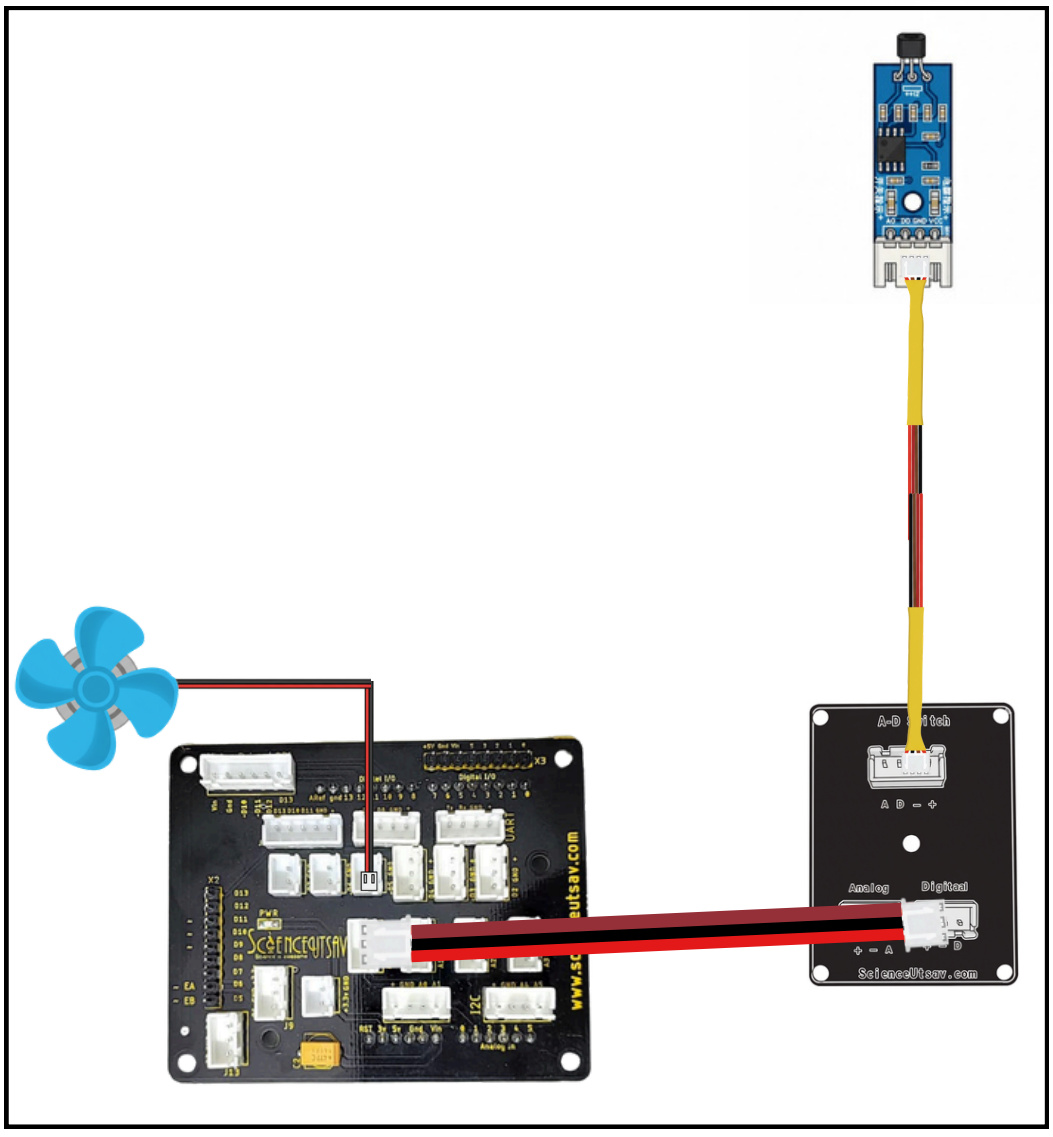


Code

```
// Session 4 Grade 8
 // Seatbelt / Headlight
 // Magnet detected = car headlight ON
 // RGB LED project: D12 is always HIGH
 // Magnetic sensor used as DIGITAL sensor on A0

 const int magneticSensor = A0;

 const int redPin = 10;
 const int greenPin = 11;
 const int alwaysHighPin = 12;

 const int MAGNET_DETECTED = LOW; // change to HIGH if opposite

 void setup() {
 pinMode(magneticSensor, INPUT);

 pinMode(redPin, OUTPUT);
 pinMode(greenPin, OUTPUT);
 pinMode(alwaysHighPin, OUTPUT);

 digitalWrite(alwaysHighPin, HIGH); // D12 always HIGH
 }

 void loop() {
 digitalWrite(alwaysHighPin, HIGH); // keep D12 always HIGH

 if (digitalRead(magneticSensor) == MAGNET_DETECTED) {
   digitalWrite(redPin, HIGH);
   digitalWrite(greenPin, HIGH);
 } else {
   digitalWrite(redPin, LOW);
   digitalWrite(greenPin, LOW);
 }
 }
```

EXPERT PROJECT

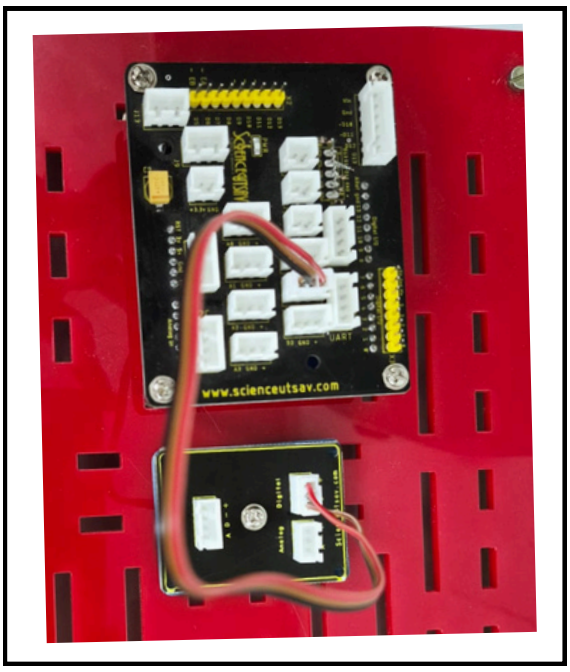
The circuit diagram of the project is given below



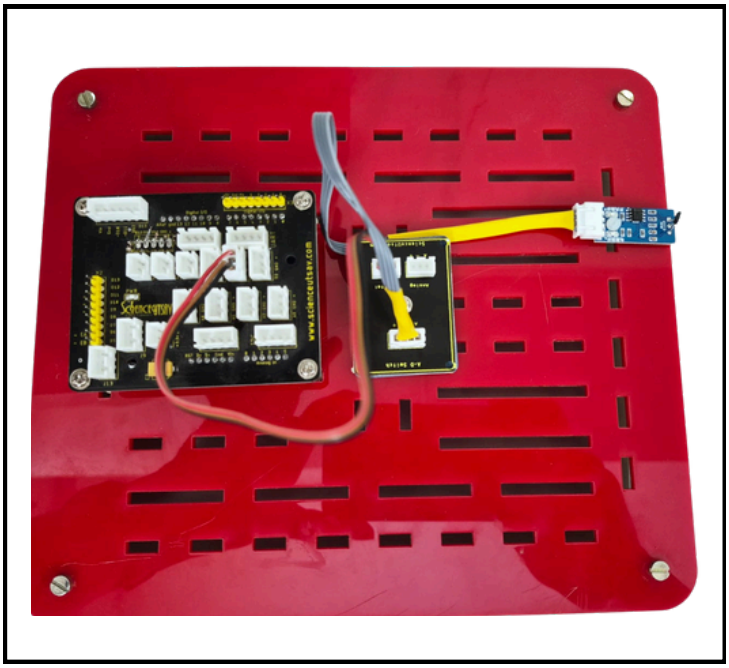
CONNECTION:

1. DC motor- D5
2. AD switch- A0

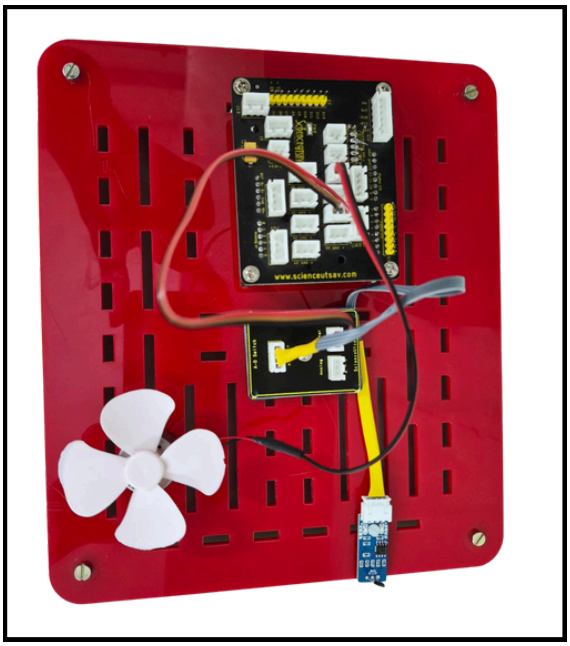
Step 1- Secure the AD switch PCB to the Pegboard



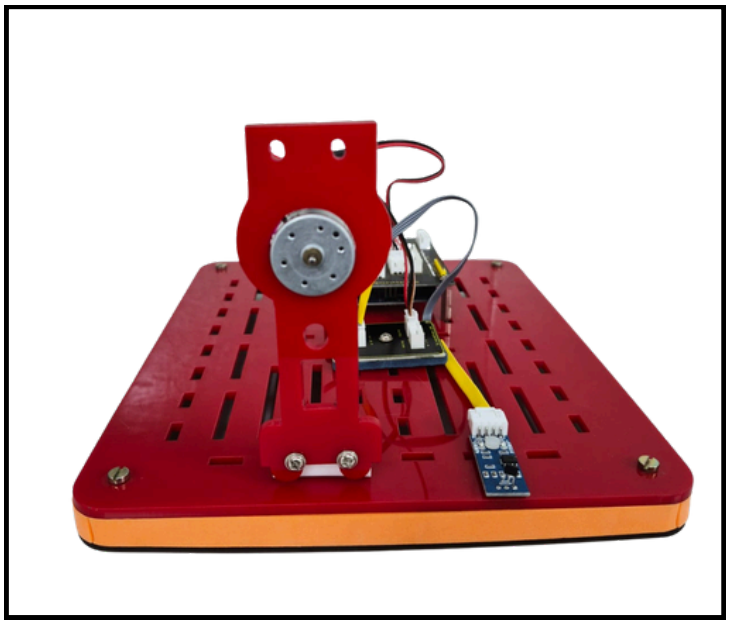
Step 2- Connect the magnetic sensor to the AD switch PCB as shown



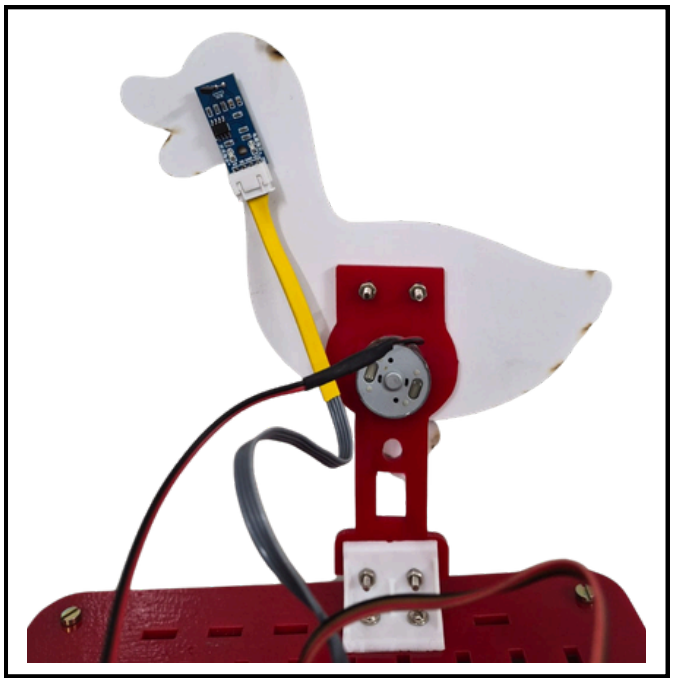
Step 3- Attach the DC motor to the shield



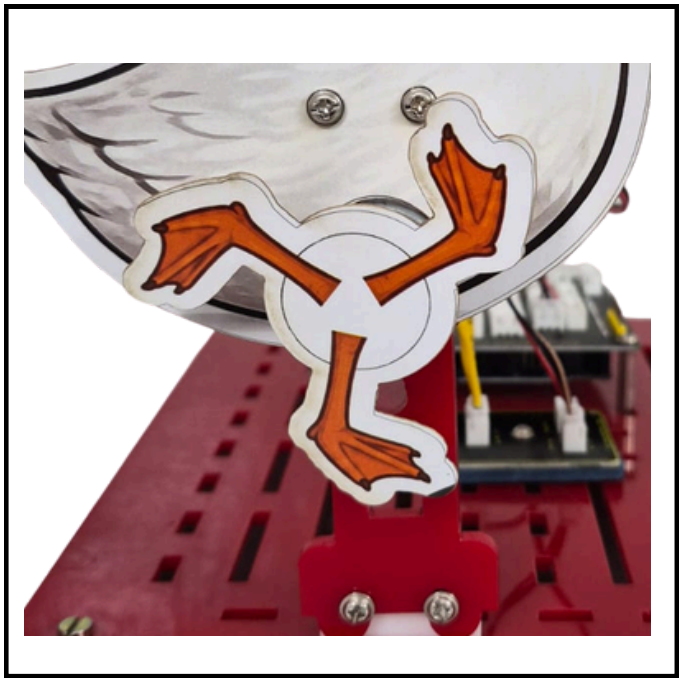
Step 4- Fix the DC motor holder to the board and attach the motor



Step 5- Attach the templte to the stand and fix the magnetic sensor to the back



Step 6- Fix the pulley to the other template and ifx it to the motor



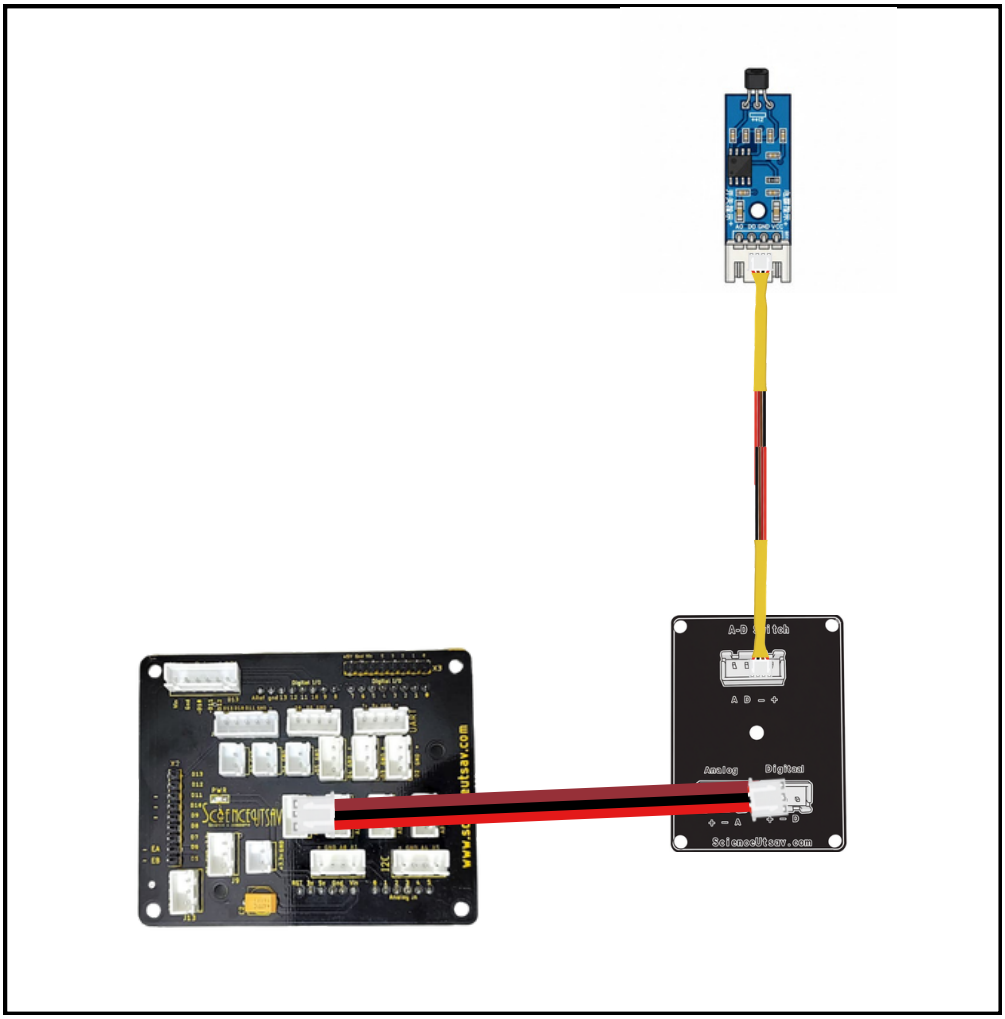
Step 7- Bring the magnet near the magnetic sensor to play with your project

Code

```
// Session 4 Grade 9
// Duck
// Leaf magnet near duck mouth sensor = duck legs motor ON
// Magnetic sensor used as DIGITAL sensor on A0

const int magneticSensor = A0;
const int motor = 5;

const int MAGNET_DETECTED = LOW;  // change to HIGH if opposite

void setup() {
  pinMode(magneticSensor, INPUT);
  pinMode(motor, OUTPUT);
}

void loop() {
  if (digitalRead(magneticSensor) == MAGNET_DETECTED) {
    digitalWrite(motor, HIGH);
  } else {
    digitalWrite(motor, LOW);
  }
}
```

EXPERT PROJECT

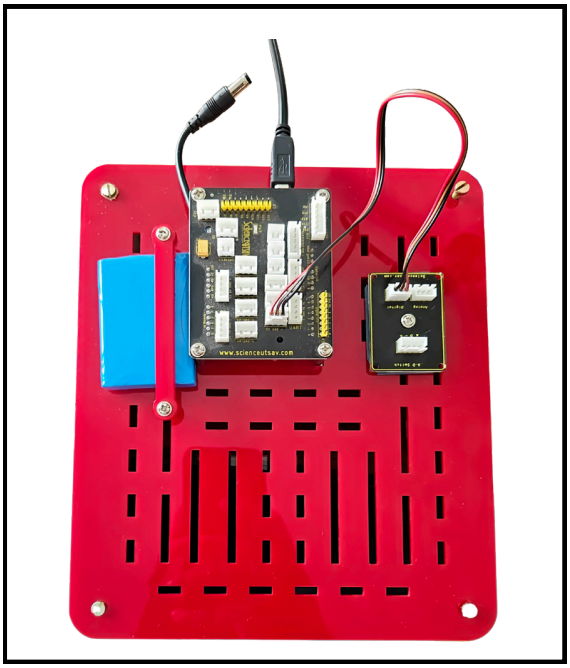
The circuit diagram of the project is given below



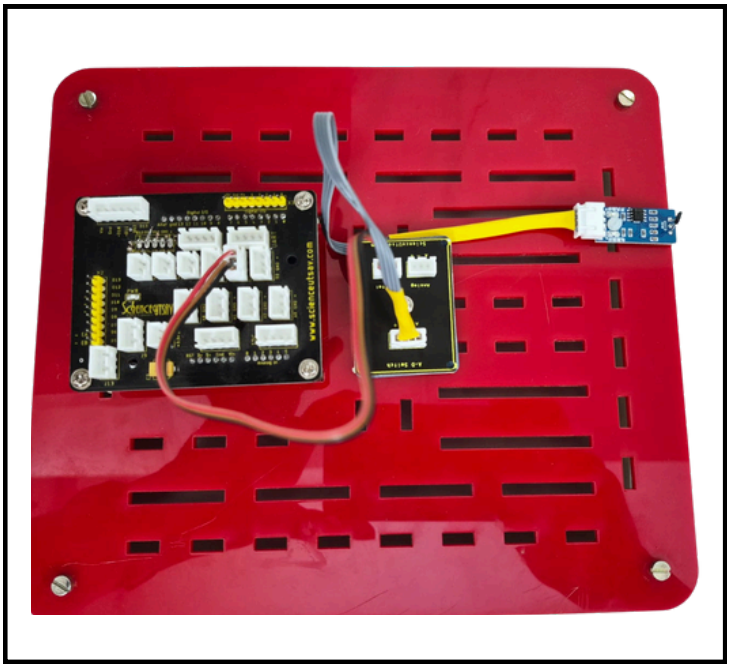
CONNECTION:

1. AD switch- A0
2. Halleffct Sensor - AD switch

Step 1- Secure the AD switch PCB to the Pegboard

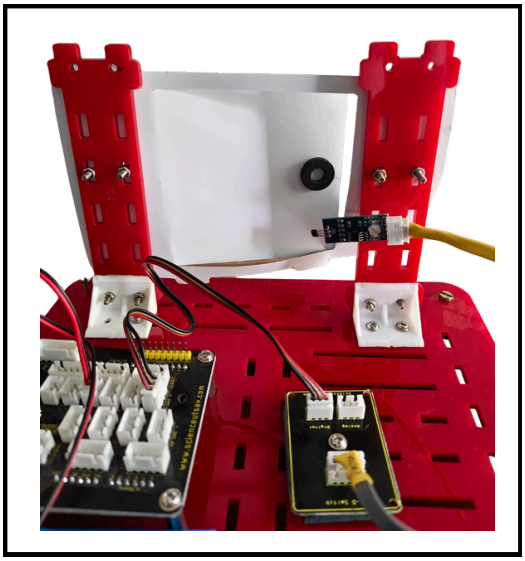


Step 2- Connect the magnetic sensor to the AD switch PCB as shown



Step 3- Place two stands and attach it to the base using L-clamps, then screw in the template to the stand

Step 4- Place and attach the magnet and halleffect sensor as shown



Step 5- Upload the code and observe

Code

```
// Session 4 Grade 9
// Magnetic Locker
// Magnet detected = door opens
// Magnetic sensor used as DIGITAL sensor on A0

#include <Servo.h>

const int magneticSensor = A0;
const int servoPin = 4;

const int MAGNET_DETECTED = LOW;  // change to HIGH if opposite

Servo lockerServo;

void setup() {
  pinMode(magneticSensor, INPUT);

  lockerServo.attach(servoPin);
  lockerServo.write(0);  // locker closed
}

void loop() {
  if (digitalRead(magneticSensor) == MAGNET_DETECTED) {
    lockerServo.write(90);  // locker open
  } else {
    lockerServo.write(0);   // locker closed
  }
}
```